

SQL Fundamentals

Q1

SELECT *
FROM employees;

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	65 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	68 000	2018-07-18	Durban
4	David	Makana	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2021-09-30	Johannes...

Q2

SELECT DISTINCT(department) AS unique_dept
FROM employees;

unique_dept
IT
HR
Finance
Marketing

Q3)

```
SELECT first_name,
       last_name
FROM employees
ORDER BY salary DESC;
```

first_name	last_name	
Eva	Naidoo	← 72000
Alice	Green	← 70000
Cathy	Zulu	← 68000
David	Moloney	← 60000
Brian	Lee	← 48000

Q4)

```
SELECT id,
       first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC
Limit 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	68000

Q5

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

FROM employees

WHERE department = 'IT';

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary
```

FROM employees

WHERE department = 'Finance'

AND salary > 60000;

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

Q7

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

FROM employees

WHERE department = 'HE'

OR department = 'Marketing';

id	first_name	last_name	department
2	Brian	Lee	HE
4	David	Mokoena	Marketing

Q8

```
SELECT id,  
       first-name,  
       last-name,  
       department
```

FROM employees

WHERE department != 'IT';

id	first-name	last-name	department
2	Brian	lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9

```
SELECT id,  
       first-name,  
       last-name,  
       department
```

FROM employees

WHERE department IN ('IT', 'HR', 'Finance');

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Q10

```
SELECT id,  
       first-name,
```

last-name,

department,

salary,

city

FROM employees

WHERE department = 'IT'

AND salary > 65000

AND city = 'Johannesburg';

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannes..
5	Eva	Naidoo	IT	72000	Johannes..